ANSWER KEY

1. (*a*)	**2.** (*d*)	**3.** (*d*)	**4.** (*d*)	**5.** (*d*)	**6.** (*b*)	**7.** (*d*)	**8.** (*c*)	**9.** (*d*)	**10.** (*a*)
11. (*b*)	**12.** (*c*)	**13.** (*d*)	**14.** (*a*)	**15.** (*d*)	**16.** (*b*)	**17.** (*b*)	**18.** (*c*)	**19.** (*b*)	**20.** (*d*)
21. (*c*)	**22.** (*a*)	**23.** [151]	**24.** [321]	**25.** [2223]	**26.** [8]	**27.** [60]	**28.** [150]	**29.** [4]	**30.** [7]

Hints and Solutions

1. (*a*) Let $x^{pq^2} = y^{qr} = z^{p^2r} = K$

$pq^2 = \log_x K$

$qr = \log_y K$

$p^2r = \log_z K$

$$\log_y x = \frac{qr}{pq^2} = \frac{r}{pq} \quad ...(i)$$

$$\log_x z = \frac{pq^2}{p^2r} = \frac{q^2}{pr} \quad ...(ii)$$

$$\log_z y = \frac{p^2r}{qr} = \frac{p^2}{q} \quad ...(iii)$$

Now, $3, \frac{3r}{pq}, \frac{3p^2}{q}, \frac{7q^2}{pr}$ in A.P

Common differenc $= \frac{3r}{pq} - 3 = \frac{1}{2}$

$$\Rightarrow r = \frac{7}{6}pq \quad ...(iv)$$

$r = pq + 1$ [Given]

$\Rightarrow pq = 6$ ...(*v*)

So, $r = 7$ $[pq + 1 = r]$ ...(*vi*)

$$\frac{3p^2}{q} = 3 + 2 \times \frac{1}{2} = 4 \quad ...(vii)$$

After solving eqn (*v*) and eqn (*vii*) $p = 2$ and $q = 3$

Hence, $r - p - q = 2$

2. (*d*) For kth AP firs term $= k$ and common differenc $= 2k - 1$

$$\left[\because S_n = \frac{n}{2}\left[2a + (n-1)d\right]\right]$$

So, $S_k = 6\,(2k + (11)\,(2k - 1))$

$S_k = 6\,(2k + 22k - 11)$

$S_k = 144k - 66$

$$\sum_{1}^{10} S_k = 144\sum_{k=1}^{10} k - 66 \times 10$$

$$= 144 \times \frac{10 \times 11}{2} - 660$$

$= 7920 - 660 = 7260$

3. (*d*) $S_1 = \frac{2n}{2}[2a + (2n-1)d] = n[2a + (2n-1)d]$ ...(*i*)

$S_2 = \frac{4n}{2}[2a + (4n-1)d] = 2n\,[2a + (4n-1)d]$ ...(*ii*)

From equ. (*ii*) – equ (*i*), we get

$S_2 - S_1 = 2n[2a + (4n-1)d] - n[2a + (2n-1)d]$

$= n[2a + (6n-1)d] = 1000$

$S_{6n} = 3n[2a + (6n-1)d] = 3000$

4. (*d*) $S_{10} = 530 \Rightarrow 5(2a + 9d) = 530$

$$2a + 9d = 106 \quad ...(i)$$

and $S_5 = 140 \Rightarrow \frac{5}{2}(2a + 4d) = 140$

$$a + 2d = 28 \quad ...(ii)$$

Solving eq (i) and eqn. (ii), we get, $a = 88$ $d = 10$

$\therefore S_{20} - S_6 = 14a + 175d = 1862$

5. (*d*) Given $3^{2\sin 2\alpha - 1}$, 14 and $3^{4 - 2\sin 2\alpha}$ are in A.P

So $3^{2\sin 2\alpha - 1} + 3^{4 - 2\sin 2\alpha} = 28$

$$\Rightarrow \frac{3^{2\sin 2\alpha}}{3} + \frac{81}{3^{2\sin 2\alpha}} = 28$$

$$\Rightarrow \frac{t}{3} + \frac{81}{t} = 28 \;\{\text{Put } 3^{2\sin 2\alpha} = t\}$$

$\Rightarrow t^2 - 84t + 243 = 0 \Rightarrow t = 81, t = 3$

$\Rightarrow$ When $t = 81$

$\Rightarrow \sin 2\alpha = 2$ (Not possible)

when $t = 3$

$\Rightarrow 2\sin 2\alpha = 1$

$\Rightarrow \sin 2\alpha = \frac{1}{2}$

$\Rightarrow 2\alpha = \frac{\pi}{6}$

$\Rightarrow \alpha = \frac{\pi}{12}$

So, first term $a = 3^0 = 1$, $d = 14 - 1 = 13$

Now, $T_6 = a + 5d = 1 + 65 = 66$

6. (*b*) Let common difference of series

$a_1, a_2, a_3, ..., a_n$ be d.

$\because a_{40} = a_1 + 39d = -159$ ... (*i*)

and $a_{100} = a_1 + 99d = -399$ ... (*ii*)

From eqn. (*ii*) and (*i*), we get

$d = -4$ and $a_1 = -3$

Hence, common difference of $b_1, b_2, b_3, ...$ is $2 - 4 = -2$.

$\therefore b_{100} = a_{70}$

$\therefore b_1 + 99(-2) = (-3) + 69(-4)$

$\therefore b_1 = 198 - 279$

$\therefore b_1 = -81$

7. (*d*) Two digit numbers of the from $7\lambda + 2$ are 16, 23 ..., 93

Two digit numbers of the from $7\lambda + 5$ are 12, 19 ..., 96

All numbers are in A.P.

So, sum of all the above number equals to

$$= \frac{12}{2}(16+93) + \frac{13}{2}(12+96) = 654 + 702 = 1356$$

$$\left[\because S_n = \frac{n}{2}\left[2a+(n-1)d\right] = \frac{n}{2}(a+l)\right]$$

8. (*c*) Let a is firs term and d is common difference

$a + 18d = 0 \Rightarrow a = -18d$

$$\frac{t_{49}}{t_{29}} = \frac{a+48d}{a+28d} = \frac{-18d+48d}{-18d+28d}$$

$$= \frac{30d}{10d} = 3$$

9. (*d*) Given, $\log_{9^{\frac{1}{2}}} x + \log_{9^{\frac{1}{3}}} x + \log_{9^{\frac{1}{4}}} x + \ldots\ldots\ldots + \log_{9^{\frac{1}{22}}} x = 504$

$\Rightarrow 2\log_9 x + 3\log_9 x + 4\log_9 x + ... + 22\log_9 x = 504$

$\Rightarrow \log_9 x + 2\log_9 x + 3\log_9 x + 4\log_9 x + ... + 22\log_9 x - \log_9 x = 504$

$\Rightarrow (1+2+3...+...+22)\log_9 x + \log_9 x = 504 \Rightarrow x = 81$

10. (*a*) $S = \sum_{i=1}^{30} a_i$, $T = \sum_{i=1}^{15} a_{2i-1}$, $a_5 = 27$, $S - 2T = 75$

Let $a_i = a + (i-1)D$

$S = a_1 + a_2 + a_3 + ... + a_{30}$

$T = a_1 + a_3 + a_5 + ... + a_{29}$

$\therefore 2T = 2a_1 + 2a_3 + 2a_5 + ... + 2a_{29}$

$S - 2T = (a_2 - a_1) + (a_4 - a_3) + (a_6 - a_5) + ... + (a_{30} - a_{29})$

$= 15\,D$

But $S - 2T = 75 \Rightarrow 15D = 75 \Rightarrow D = 5$

Now $a_5 = 27 \Rightarrow a + 4D = 27$

$\therefore a = 27 - 20 \Rightarrow a = 7$

Now $a_{10} = a + 9D$

$= 7 + 45 = 52$

11. (*b*)

12. (*c*) $\because \alpha, \beta, \gamma, \delta$ are in G.P, so $\alpha\delta = \beta\gamma$

$$\Rightarrow \frac{\alpha}{\beta} = \frac{\gamma}{\delta} \Rightarrow \left|\frac{\alpha-\beta}{\alpha+\beta}\right| = \left|\frac{\gamma-\delta}{\gamma+\delta}\right|$$

$$\Rightarrow \left|\frac{\sqrt{(\alpha+\beta)^2 - 4\alpha\beta}}{\alpha+\beta}\right| = \left|\frac{\sqrt{(\gamma+\delta)^2 - 4\gamma\delta}}{\gamma+\delta}\right|$$

$$\Rightarrow \frac{\sqrt{9-4p}}{3} = \frac{\sqrt{36-4q}}{6}$$

$\Rightarrow 36 - 16p = 36 - 4q$

$\Rightarrow q = 4p$

So, $\frac{2q+p}{2q-p} = \frac{9p}{7p} = \frac{9}{7}$

13. (*d*) $(a^2p^2 - 2abp + b^2) + (b^2p^2 - 2bcp + c^2) + (c^2p^2 - 2dcp + d^2) = 0$

$\Rightarrow (ap - b)^2 + (bp - c)^2 + (cp - d)^2 = 0$

$\therefore \quad ap - b = bp - c = cp - d = 0$

$ap = b, bp = c, cp = d$

$\Rightarrow \frac{b}{a} = \frac{c}{b} = \frac{d}{c} = p \therefore a, b, c, d$ are in G.P.

14. (*a*) $\alpha x^2 + 2\beta x + \gamma = 0$

Let $\beta = \alpha t, \gamma = \alpha t^2$

$\therefore \alpha x^2 + 2\alpha tx + \alpha t^2 = 0$

$\Rightarrow x^2 + 2tx + t^2 = 0$

$\Rightarrow (x + t)^2 = 0 \Rightarrow x = -t$

It must be root of equation $x^2 + x - 1 = 0$

$\therefore t^2 - t - 1 = 0$ …(*i*)

Now

$\alpha(\beta+\gamma) = \alpha^2 (t + t^2)$

Option (a) $\beta\gamma = \alpha t \,.\, \alpha t^2 = \alpha^2 t^3 = \alpha^2 (t^2 + t)$

$[\because t^3 = t^2 + t]$

from equation (i)

15. (d) $a, ar, ar^2, \ldots$

$$T_2 + T_6 = \frac{25}{2} \Rightarrow ar\left(1+r^4\right) = \frac{25}{2}$$

$$a^2r^2\left(1+r^4\right)^2 = \frac{625}{4} \quad \ldots (1)$$

$$T_3 \cdot T_5 = 25 \Rightarrow \left(ar^2\right)\left(ar^4\right) = 25$$

$$a^2r^6 = 25 \quad .. (2)$$

On dividing by (2)

$$\frac{\left(1+r^4\right)^2}{r^4} = \frac{25}{4} \Rightarrow 4r^8 - 17r^4 + 4 = 0$$

$$(4r^4 - 1)(r^4 - 4) = 0$$

$$r^4 = \frac{1}{4}, 4 \Rightarrow r^4 = 4 \text{ (an increasing geometric series)}$$

$$a^2r^6 = 25 \Rightarrow \left(ar^3\right)^2 = 25$$

$$T_4 + T_6 + T_8 = ar^3 + ar^5 + ar^7$$

$$a = ar^3\left(1+r^2+r^4\right) = 5(1+2+4) = 35$$

16. (b) $T_r = \dfrac{\left(r^2+r+1\right)-\left(r^2-r+1\right)}{2\left(r^4+r^2+1\right)}$

we know,

$r^4 + r^2 + 1 = (r^2 + r + 1)(r^2 - r + 1)$

$$\Rightarrow T_r = \frac{1}{2}\left[\frac{1}{r^2-r+1} - \frac{1}{r^2+r+1}\right]$$

$$T_1 = \frac{1}{2}\left[\frac{1}{1} - \frac{1}{3}\right]$$

$$T_2 = \frac{1}{2}\left[\frac{1}{3} - \frac{1}{7}\right]$$

$$T_3 = \frac{1}{2}\left[\frac{1}{7} - \frac{1}{13}\right]$$

$$\vdots$$

$$T_{10} = \frac{1}{2}\left[\frac{1}{91} - \frac{1}{111}\right]$$

$$\Rightarrow \sum_{r=1}^{10} T_r = \frac{1}{2}\left[1 - \frac{1}{111}\right] = \frac{55}{111}$$

17. (b) a, A_1, A_2, b are in $A.P.$

$$d = \frac{b-a}{3}; A_1 = a + \frac{b-a}{3} = \frac{2a+b}{3}$$

$$A_2 = \frac{a+2b}{3}$$

$A_1 + A_2 = a + b$

a, G_1, G_2, G_3, b are in $G.P.$

$$r = \left(\frac{b}{a}\right)^{\frac{1}{4}}$$

$$G_1 = \left(a^3b\right)^{\frac{1}{4}}$$

$$G_2 = \left(a^2b^2\right)^{\frac{1}{4}}$$

$$G_3 = \left(ab^3\right)^{\frac{1}{4}}$$

$$G_1^4 + G_2^4 + G_3^4 + G_1^2 G_2^2 =$$

$$a^3b + a^2b^2 + ab^3 + \left(a^3b\right)^{\frac{1}{2}} \cdot \left(ab^3\right)^{\frac{1}{2}}$$

$= a^3b + a^2b^2 + ab^3 + a^2.b^2$

$= ab(a^2 + 2ab + b^2)$

$= ab(a + b)^2$

$= G_1.G_3.(A_1 + A_2)^2$

18. (c) $S_n = 4 + 11 + 21 + 34 \ldots.\ T_n$

$S_n = \quad 4 + 11 + 21 \ldots.\ T_{n-1} + T_n$

$\Rightarrow 0 = (4 + 7 + 10 + 13 + \ldots) - T_n$

$\Rightarrow T_n = 4 + 7 + 10 + 13 + \ldots$

$$= \frac{n}{2}[8 + (n-1)3]$$

$$T_n = \frac{n}{2}[3n+5]$$

$$S_n = \sum T_n$$

$$= \frac{3}{2}\sum n^2 + \frac{5}{2}\sum n$$

$$= \frac{3}{2}\frac{n(n+1)(2n+1)}{6} = \frac{5}{2}\frac{(n)(n+1)}{2}$$

$$= \frac{n(n+1)}{4}[2n+1+5]$$

$$S_n = \frac{n(n+1)}{4}(2n+6) = \frac{n(n+1)(n+3)}{2}$$

Now, $\dfrac{1}{60}(S_{29} - S_9)$

$$= \frac{1}{60}\left(\frac{29\times30\times32}{2} - \frac{9\times10\times12}{2}\right) = 223$$

19. (*b*) $1^2-2^2+3^2-4^2+....(2021)^2-(2022)^2+(2023)^2=1012\,m^2n$

$\Rightarrow (1-2)(1+2)+(3-4)(3+4)+...+(2021-2022)$
$(2021+2022)+(2023)^2=1012\,m^2n$

$\Rightarrow (-1)(1+2+3+4+...+2022)+(2023)^2=1012\,m^2n$

$\Rightarrow (-1)\cdot\dfrac{(2022)(2023)}{2}+(2023)^2=1012\,m^2n$

$\Rightarrow 2023(2023-1011)=1012\,m^2n$

$\Rightarrow 2023\times 1012=1012\,m^2n$

$\Rightarrow m^2n=2023=17^2.7$

$\therefore\ m=17, n=7$

Hence, $m^2-n^2=17^2-7^2=240$

20. (*d*) $x_1y>0$ and $x^3y^2=2^{15}$

AM $\geq$ GM

$\dfrac{3x+2y}{5}\geq\left(x^3y^2\right)^{\frac{1}{5}}$

$\Rightarrow 3x+2y\geq 40$

21. (*c*) $\dfrac{x^2}{2}+\dfrac{x^2}{2}+\dfrac{x^2}{2}+\dfrac{x^2}{2}+\dfrac{x^2}{2}+\dfrac{\alpha}{2x^5}+\dfrac{\alpha}{2x^5}\geq 7\left(\dfrac{\alpha^2}{2^7}\right)^{\frac{1}{7}}$

$\Rightarrow \dfrac{7\cdot(\alpha)^{2/7}}{2}=14$

$\Rightarrow (\alpha^2)^{1/7}=2^2$

$\Rightarrow \alpha=(2^2)^{7/2}=2^7$

Therefore, $\alpha=128$

22. (*a*) $S=\log_7x^2+\log_7x^3+\log_7x^4+...$ 20 terms

$\Rightarrow \log_7(x^2\cdot x^3\cdot x^4\cdot\ ...\ x^{21})=460$ [$\because$ Given]

$\Rightarrow \log_7x^{(2+3+4...21)}=460$

$\Rightarrow (2+3+4+...+21)\log_7 x=460$

$\Rightarrow \dfrac{20}{2}(2+21)\log_7x=460$

$\Rightarrow \log_7 x=\dfrac{460}{230}=2\ \Rightarrow x=7^2=49$

23. [151] $d=\text{LCM}(d_1,d_2)=\text{LCM}(4,5)=20$

Series of common term is 11, 31, 51, 71....

8th common term in both the series is

$T_8=11+(8-1)\times 20$

$=11+140=151$

24. [321] $3,7,11,15,\ldots\ldots\ldots\ldots,399\ d_1=4$

$2,5,8,11,\ldots\ldots\ldots\ldots,359\ d\ =3$

$2,7,12,17,\ldots\ldots,197\ d_3=5$

$LCM\left(d_1,d_2,d_3\right)=60$

By hit and trial guess firs common term

Common terms are 47, 107, 167

Sum = 321

25. [2223] First common term to both AP's is 9 and common difference is 12.

t_{78} of $(3,6,9,....)=78\times 3=234$

t_{59} of $(5,9,13,....)=5+(59-1)4=237$

n^{th} common term ≤ 234

$9+(n-1)12\leq 234$

$n<\dfrac{237}{12}\Rightarrow n=19$

Now sum of 19 terms with $a=9$, $d=12$

$=\dfrac{19}{2}\left(2.9+(19-1)\cdot 12\right)=2223$

26. [8] $2a_7=a_5$ (given)

$\Rightarrow 2(a_1+6d)=a_1+4d$

$\Rightarrow a_1+8d=0$ ...(*i*)

and given $a_{11}=18\Rightarrow\ a_1+10d=18$ ...(*ii*)

By (*i*) and (*ii*) we get $a_1=-72$, $d=9$

$a_{18}=a_1+17d=-72+153=81$

$a_{10}=a_1+9d=9$

Now

$12\left(\dfrac{\sqrt{a_{11}}-\sqrt{a_{10}}}{d}+\dfrac{\sqrt{a_{12}}-\sqrt{a_{11}}}{d}+.....\dfrac{\sqrt{a_{18}}-\sqrt{a_{17}}}{d}\right)$

$=12\left(\dfrac{\sqrt{a_{18}}-\sqrt{a_{10}}}{d}\right)=\dfrac{12(9-3)}{9}=\dfrac{12\times 6}{9}=8$

27. [60] $a_4\cdot a_6=9\Rightarrow\left(a_5\right)^2=9\Rightarrow a_5=3$

$\&\ a_5+a_7=24\Rightarrow a_5+a_5r^2=24$

$\Rightarrow\left(1+r^2\right)=8\Rightarrow r=\sqrt{7}$

$\because\ a_5=3\Rightarrow ar^4=3$

$\Rightarrow a=\dfrac{3}{49}$

$\Rightarrow a_1a_9+a_2a_4a_9+a_5+a_7=9+27+3+21=60$

28. [150] Since, $\dfrac{1}{x},\dfrac{1}{y},\dfrac{1}{z}$ are in AP

$\Rightarrow \dfrac{2}{y}=\dfrac{1}{x}+\dfrac{1}{z}$

$\because x, \sqrt{2}y, z$ are in G.P.

$\left(\sqrt{2}y\right)^2 = xz \Rightarrow 2y^2 = xz$

$\therefore \frac{2}{y} = \frac{x+z}{xz} \Rightarrow \frac{2}{y} = \frac{x+z}{2y^2}$

$\Rightarrow x + z = 4y$

$\because xy + yz + zx = \frac{3}{\sqrt{2}} xyz$

$y(x+z) + zx = \frac{3}{\sqrt{2}} xz \cdot y$

$\Rightarrow 4y^2 + 2y^2 = \frac{3}{\sqrt{2}} \times 2y^2 \times y$

$\Rightarrow 6y^2 = 3\sqrt{2}y^3 \Rightarrow y = \sqrt{2}$

$\therefore x + y + z = 5y = 5\sqrt{2}$

Hence, $3(x + y + z)^2 = 3 \times 50 = 150$

29. [4] $S = 1 + 3 + 3^2 + 3^3 + \ldots + 3^{2021}$

$= \frac{3^{2022} - 1}{2} = \frac{1}{2}\left[a^{1011} - 1\right]$

$= \frac{1}{2}\left[9^{1011} - 1\right] = \frac{1}{2}[100k + 10110 - 1 - 1]$

$= 50k_1 + 4$

Remainder = 4

30. [7] $P = \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{2^2} - \frac{1}{2.3} + \frac{1}{3^2}\right) +$

Given expression is $(x - y) + (x^2 - xy + y^2) + (x^2 - x^2y + xy^2 - y^3) + \ldots$

On multiplying $(x + y)$, we get

$= (x^2 - y^2) + (x^3 + y^2) + (x^4 - y^4) + \ldots$

$= (x^2 + x^3 + x^4 + \ldots) - (y^2 - y^3 + y^4 - \ldots)$

$= \frac{x^2}{1-x} - \frac{y^2}{1+y}$

$\therefore P\left(\frac{1}{2} + \frac{1}{3}\right) = \left(\frac{1}{2^2} - \frac{1}{3^2}\right) + \left(\frac{1}{2^3} + \frac{1}{3^3}\right) + \left(\frac{1}{2^4} - \frac{1}{3^4}\right) + \ldots$

$\frac{5P}{6} = \frac{\frac{1}{4}}{1 - \frac{1}{2}} - \frac{\frac{1}{9}}{1 + \frac{1}{3}}$

$\Rightarrow \frac{5P}{6} = \frac{1}{2} - \frac{1}{12} = \frac{5}{12}$

$\therefore P = \frac{1}{2} = \frac{\alpha}{\beta}$

Hence, $\alpha = 1, \beta = 2$

$\Rightarrow \alpha + 3\beta = 7$